

EEAIC assignment5

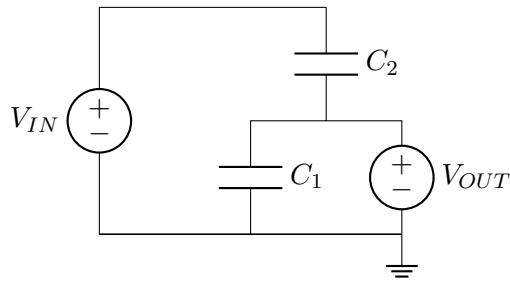
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Part I - Modeling of the Dickson and Series-Parallel Switched-Capacitor Step-Down Converter

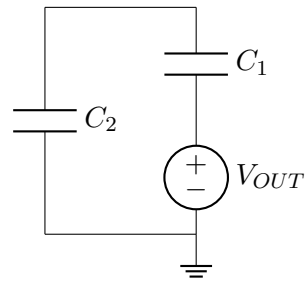
Parameter	Value	Parameter	Value
V_{IN}	3.0 V	k_{bot}	0.005
V_{OUT} (nominal)	1.0 V	f_{SW}	100 MHz
C	10 nF	$R_{DS\,ON}$	500 m Ω
C_{fly}	1 nF		

1. Draw circuit in phase 1 and phase 2

Dickson

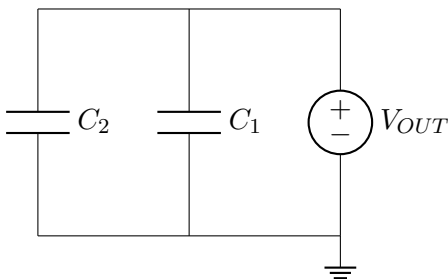


Phase I

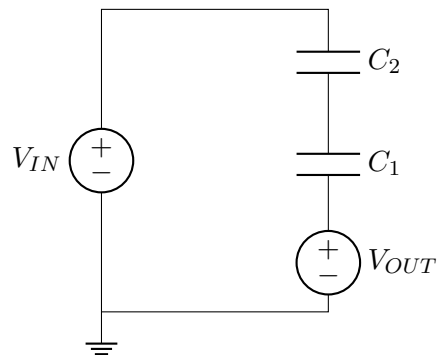


Phase II

Series-Parallel



Phase I



Phase II

2. Capacitor Balancing Voltages

By applying KVL to each phase of the converters, we can derive the balanced DC voltages of the flying capacitors. We expect these voltages to remain effectively constant between phases, as the converters typically operates in a regime where the total stored charge is significantly larger than the charge delivered to the load in a single cycle, thereby suppressing voltage fluctuations.

Dickson:

$$\text{Phase 1: } V_{IN} - V_{C2} = V_{OUT}$$

$$V_{C1} = V_{OUT}$$

$$\text{Phase 2: } V_{C2} - V_{C1} = V_{OUT}$$

Therefore:

$$V_{C2} = 2V_{OUT}$$

$$V_{IN} = 3V_{OUT}$$

Corresponding to the following numerical values:

$$V_{C1} = V_{OUT} = 1 \text{ V}$$

$$V_{C2} = V_{IN} - V_{OUT} = 2 \text{ V}$$

Series-Parallel:

$$\text{Phase 1: } V_{C1} = V_{C2} = V_{OUT} = 1 \text{ V}$$

$$\text{Phase 2: } V_{IN} - V_{C2} - V_{C1} = V_{OUT}$$

Therefore:

$$V_{OUT} = \frac{V_{IN}}{3}$$

These steady-state relations demonstrate that these converter configurations inherently force a 3:1 conversion ratio between the input and the output.

3. Charge Multiplier Vectors (R_{SSL})

For the slow-switching limit (SSL) impedance analysis, the finite resistances of the switches, capacitors, and interconnect are neglected. A pair of charge multiplier vectors a^1 and a^2 can be derived for any standard non-degenerate two-phase SC converter. The charge multiplier vectors correspond to charge flows that occur immediately after the switches are closed to initiate each respective phase of the SC circuit. Each element of a charge multiplier vector corresponds to a specific capacitor or independent voltage source, and represents the charge flow into that component, normalized with respect to the output charge flow.

The charge multiplier vector a_c defines the normalized charge flow through each capacitor. By applying KCL and ensuring $q_{C,i}^1 = -q_{C,i}^2$:

$$a^1 = \begin{bmatrix} q_{OUT}^{(1)} & q_1^{(1)} & q_2^{(1)} & q_{IN}^{(1)} \end{bmatrix}^T / q_{OUT}$$

$$a^2 = \begin{bmatrix} q_{OUT}^{(2)} & q_1^{(2)} & q_2^{(2)} & q_{IN}^{(2)} \end{bmatrix}^T / q_{OUT}$$

Dickson:

$$\text{Phase 1: } q_{OUT}^{(1)} = q_1 + q_2$$

$$\text{Phase 2: } q_{OUT}^{(2)} = q_1 = q_2$$

Therefore:

$$q_{OUT} = q_{OUT}^{(1)} + q_{OUT}^{(2)} = 3q_1 = 1$$

$$\Rightarrow q_1 = \frac{1}{3} = q_2$$

$$\begin{bmatrix} q_{OUT}^{(1)} & q_1^{(1)} & q_2^{(1)} & q_{IN}^{(1)} \end{bmatrix} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & \frac{1}{3} & -\frac{1}{3} \end{bmatrix}$$

$$\begin{bmatrix} q_{OUT}^{(2)} & q_1^{(2)} & q_2^{(2)} & q_{IN}^{(2)} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & -\frac{1}{3} & 0 \end{bmatrix}$$

Series-Parallel:

Also in this case we note that:

$$\text{Phase 1: } q_{OUT}^{(1)} = q_1 + q_2$$

$$\text{Phase 2: } q_{OUT}^{(2)} = q_1 = q_2$$

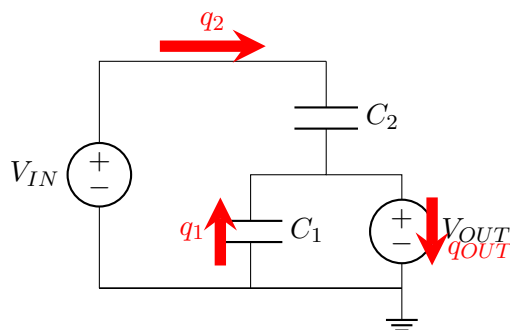
Therefore we extract the same result of the Dickson case:

$$\begin{bmatrix} q_{OUT}^{(1)} & q_1^{(1)} & q_2^{(1)} & q_{IN}^{(1)} \end{bmatrix} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & \frac{1}{3} & -\frac{1}{3} \end{bmatrix}$$

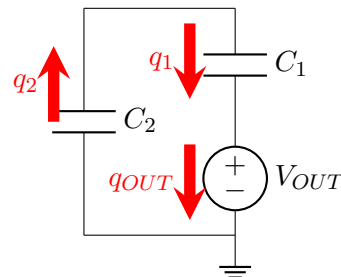
$$\begin{bmatrix} q_{OUT}^{(2)} & q_1^{(2)} & q_2^{(2)} & q_{IN}^{(2)} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & -\frac{1}{3} & 0 \end{bmatrix}$$

You can observe that the sum of the output charge multiplier components is equal to 1 ($\frac{2}{3} + \frac{1}{3}$) and that each charge multiplier component of the flying capacitor in Phase I has opposite sign and same module than in Phase II.

For both 3:1 step-down converters, to transfer 1 unit of charge to the output, $\frac{1}{3}$ unit of charge must be drawn from the input.

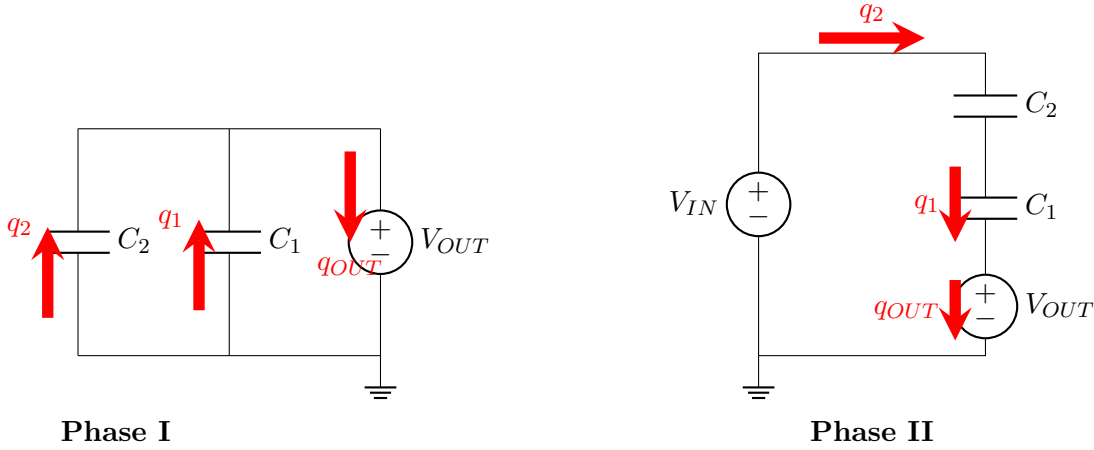
Dickson

Phase I



Phase II

Series-Parallel



4. Slow Switching Limit R_{SSL}

The R_{SSL} measures the impedance assuming switches are ideal and capacitors fully charge/discharge. Using the formula $R_{SSL} = \sum_i \frac{(a_{c,i})^2}{C_i f_{sw}}$:

$$R_{SSL} = \frac{1}{C_{fly} f_{sw}} \left[\left(\frac{1}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right] \approx 2.22 \Omega$$

This value holds for both topologies since their a_c vectors and capacitor values are identical.

5. Charge Multiplier Vector (R_{FSL})

The other asymptotic limit, the fast switching limit (FSL), is characterized by nearly-constant current flows between capacitors. The switch on-state impedances and other resistances are sufficiently large such that during each phase, the capacitors do not approach equilibrium. In the asymptotic limit, the capacitor voltages are modeled as constant. The circuit loss is related only to conduction loss in resistive elements.

The duty cycle of the converter is important when considering the FSL impedance since currents flow during the entirety of each phase. For this analysis, a duty cycle of 50% is assumed for simplicity.

Additionally, only the on-state switch resistance is considered; other parasitic resistance (i.e. capacitor equivalent series resistance (ESR)) can be similarly incorporated into the model.

The calculation of the fast switching asymptotic limit follows a similar procedure. We need to estimate one charge multiplier vector defined as the charge flow through each switch during the phase in which the switch is closed.

For both the 3:1 Dickson and Series-Parallel converters, which utilize 7 switches, all switches conduct an identical portion of the charge relative to the output:

$$a_r = [q_{SW1} \ q_{SW2} \ q_{SW3} \ q_{SW4} \ q_{SW5} \ q_{SW6} \ q_{SW7}] = \left[\frac{1}{3} \ \frac{1}{3} \ \frac{1}{3} \ \frac{1}{3} \ \frac{1}{3} \ \frac{1}{3} \ \frac{1}{3} \right]$$

6. Fast Switching Limit R_{FSL}

The R_{FSL} evaluates conduction losses assuming capacitors are fixed voltage sources. Utilizing $R_{FSL} = 2 \sum_i R_{DS(on),i} (a_{r,i})^2$:

$$R_{FSL} = 2R_{DS(on)} \left[7 \times \left(\frac{1}{3} \right)^2 \right] = 2 \cdot 0.5 \cdot \left(\frac{7}{9} \right) \approx 0.778 \Omega$$

7. Converter Output Resistance

The total output resistance merges the two asymptotic limits:

$$R_{OUT} \simeq \sqrt{R_{SSL}^2 + R_{FSL}^2} = \sqrt{(2.22)^2 + (0.778)^2} \approx 2.35 \Omega$$

8. Topology Comparison

- **Conduction Losses:** The Dickson and Series-Parallel topologies demonstrate superior conduction efficiency compared to the Ladder configuration. Specifically, they achieve R_{SSL} and R_{FSL} coefficients of 2/9 and 14/9, significantly lower than the Ladder's values of 2/3 and 8/3. In practical terms, this results in a lower R_{FSL} (0.778 Ω vs 1.33 Ω), which minimizes power dissipation through the MOSFETs at high frequencies. By simultaneously reducing the R_{SSL} coefficient to 2/9, these topologies optimize charge transfer and minimize the energy loss inherent in the periodic "pumping" of the flying capacitors. Since the total output resistance R_{OUT} is a combination of these two asymptotic limits, the Dickson and Series-Parallel converters provide a smaller overall impedance, which is fundamental to maximizing conversion efficiency.

- **Switching Losses:**

Switching losses are dominated by the energy required to charge parasitic capacitances ($P_{SW} = C \cdot f_{sw} \cdot V_{swing}^2$).

- **Bottom plate capacitance losses:** For the Ladder configuration, only C_2 and C_4 contribute significantly with a 1 V swing, while C_3 is tied to V_{OUT} , resulting in $P_{SW,bott,Ladder} = C_{bott} f_{sw} (V_{OUT}^2 + 1^2)$. The Dickson and Series-Parallel topologies use only two capacitors. In Dickson, both C_1 and C_2 bottom plates swing by V_{OUT} , yielding $P_{SW,bott,Dickson} = 2C_{bott} f_{sw} V_{OUT}^2$. The Series-Parallel converter suffers the highest losses, as C_2 swings between ground and the 2 V potential of C_1 top, resulting in $P_{SW,bott,SP} = C_{bott} f_{sw} (V_{OUT}^2 + 2^2)$.
- **Drain switching losses:** In the Ladder topology, SW3 and SW5 experience 1 V swings and SW1 swings by V_{OUT} . The Dickson converter presents a more complex distribution: SW1 and SW4 swing by V_{OUT} , while SW5 and SW6 swing by 1 V. The Series-Parallel configuration again exhibits the highest losses; while SW1 swings by V_{OUT} , both SW4 and SW6 undergo 2 V excursions due to their connections between ground - 2 V nodes and V_{IN} - V_{OUT} respectively.
- **Gate switching losses:** With $C_{gate} \approx 3.1$ pF according to our testbench and a similar average $V_{swing,gate}^2 \approx 5.5$ V² across all topologies, the total loss is primarily dictated by the switch count. Although the Dickson converter shows the highest per-switch voltage stress, the Ladder topology achieves the lowest overall gate losses because it utilizes only 6 switches, compared to the 7 required by both the Dickson and Series-Parallel configurations.
- **Component Count:** To achieve a 3:1 ratio, the Ladder requires 3 capacitors and 6 switches. Dickson and Series-Parallel require only 2 capacitors but 7 switches.

- **Flying Capacitors and Switching utilization:** The Ladder converter is the most efficient regarding device ratings, as every switch and capacitor is uniformly stressed to only 1 V. This allows for compact, low-voltage components that minimize both silicon area and parasitic overhead. In contrast, the Dickson topology requires capacitor C_2 and switch SW6 to withstand 2 V, necessitating larger, higher-voltage components with increased parasitic capacitances. Similarly, while the Series-Parallel configuration maintains a 1 V balance across all capacitors, it subjects switches SW5, SW6, and SW7 to 2 V V_{DS} stresses. These transistors require significantly larger gate areas to maintain the same on-resistance, increasing gate capacitance and degrading high-frequency performance relative to the Ladder topology.

The comparison of the three topologies reveals a fundamental trade-off between conduction efficiency and parasitic-induced switching losses. The **Dickson** and **Series-Parallel** configurations are superior in terms of conduction resistance due to their lower R_{SSL} and R_{FSL} coefficients ($2/9$ and $14/9$), making them ideal for low-frequency applications or designs where minimizing output impedance is the priority. Additionally, they achieve the 3:1 ratio with only two capacitors, potentially reducing the passive component footprint.

Conversely, the **Ladder** topology excels in switching loss and voltage stress management. By maintaining a uniform 1 V stress across all devices, it avoids the parasitic overhead of higher-voltage components. While more resistive theoretically, its reduced component count and lower parasitic voltage swings make it highly efficient at high frequencies where dynamic power dominates. In summary, the **Series-Parallel** topology maximizes conduction efficiency at the cost of high parasitic swings, the **Dickson** provides a middle ground with moderate stress, and the **Ladder** offers the most benign operating environment and the lowest switching power. The optimal selection ultimately depends on the target switching frequency and the specific constraints of the process technology.

Part II - Simulation of the Dickson SC Converter on Cadence

1. Dickson Testbench

The switches are implemented using $nch-25$ NMOS transistors from the TSMC N65 library. To ensure optimal performance, the gate-source voltage (V_{GS}) is configured to swing between 0 V for the OFF state and 1.8 V for the ON state and the channel length is fixed at the technology minimum L_{min} .

To determine the transistor dimensions required to achieve a target ON-resistance (R_{DSon}) of $500\text{ m}\Omega$, a parametric sweep of the transistor width (W) was performed in a dedicated characterization testbench, similarly to *Assignment 4*.

To achieve the target resistance, based on the simulation results, is required a total transistor width of:

$$W_{500\text{m}\Omega, R_{DSon}} = 2.717\text{ mm}$$

The complete schematic is presented in *Figure 1*, with detailed views of the upper and lower circuits provided in *Figure 2* and *Figure 3*, respectively. These figures illustrate the final transistor sizing and voltage source configurations implemented to satisfy the specified design requirements

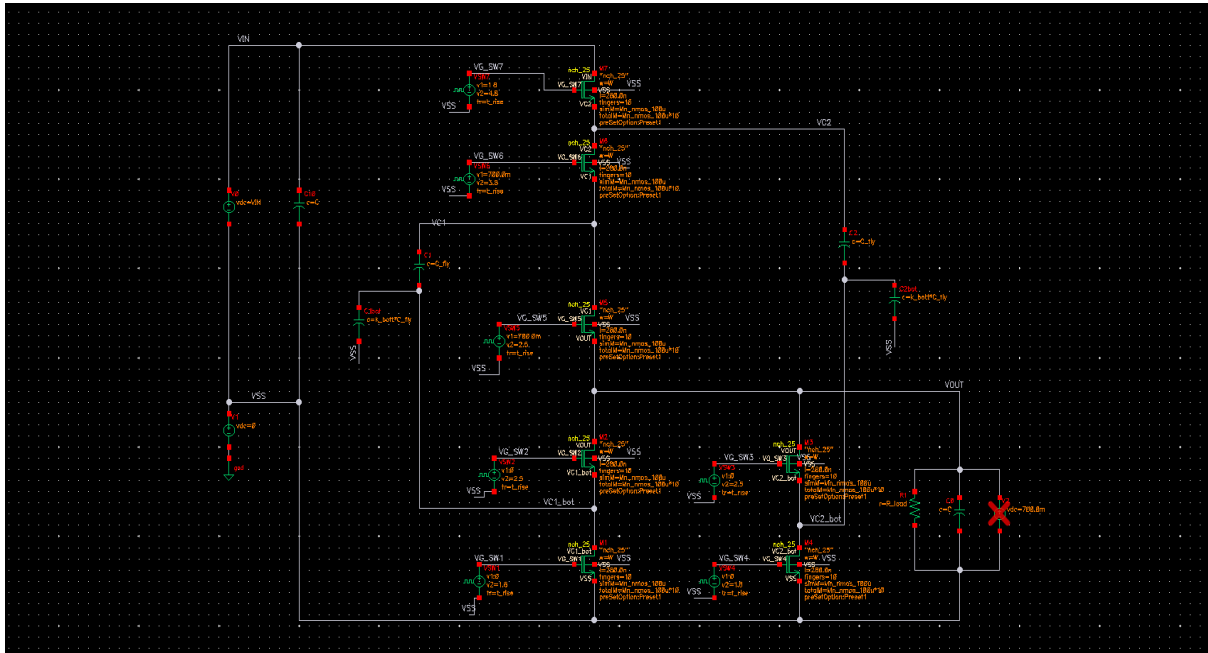


Figure 1: Dickson converter schematic

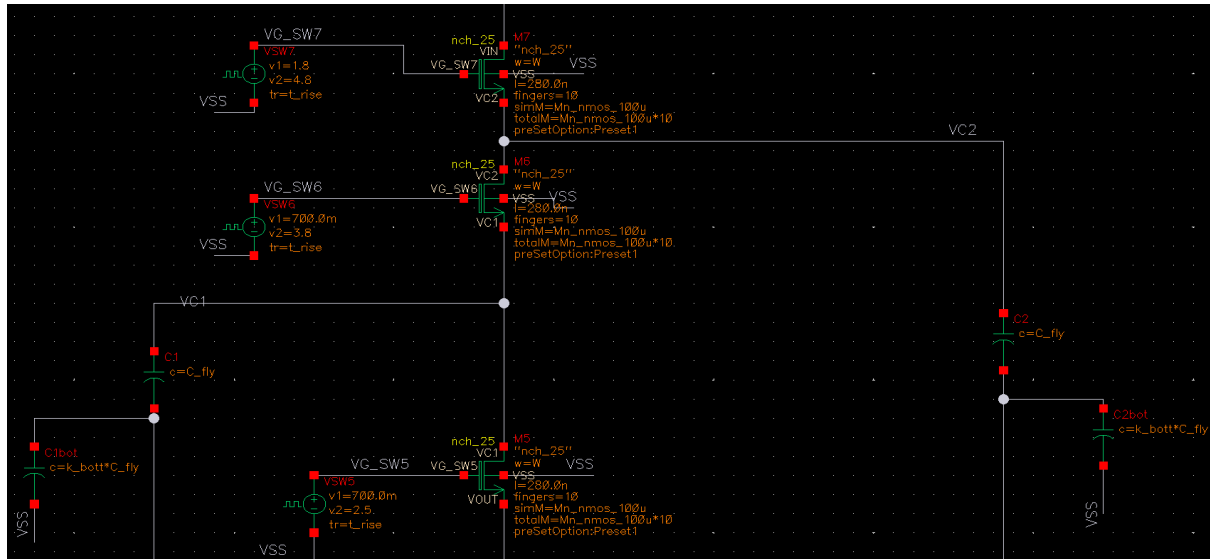


Figure 2: Zoom on upper schematic of Dickson converter

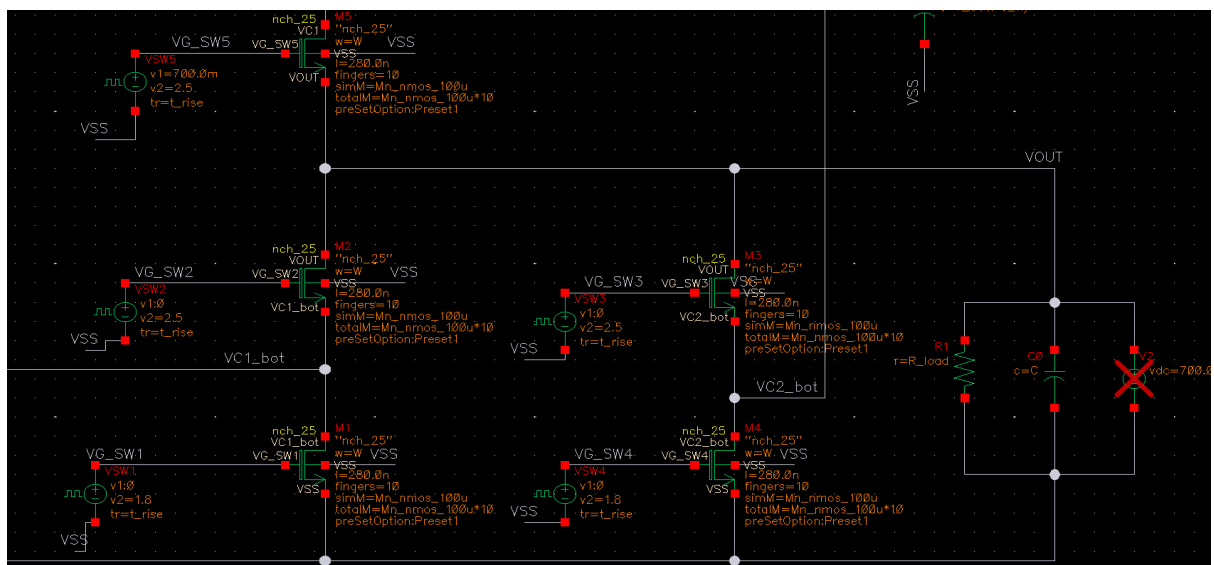


Figure 3: Zoom on lower schematic of Dickson converter

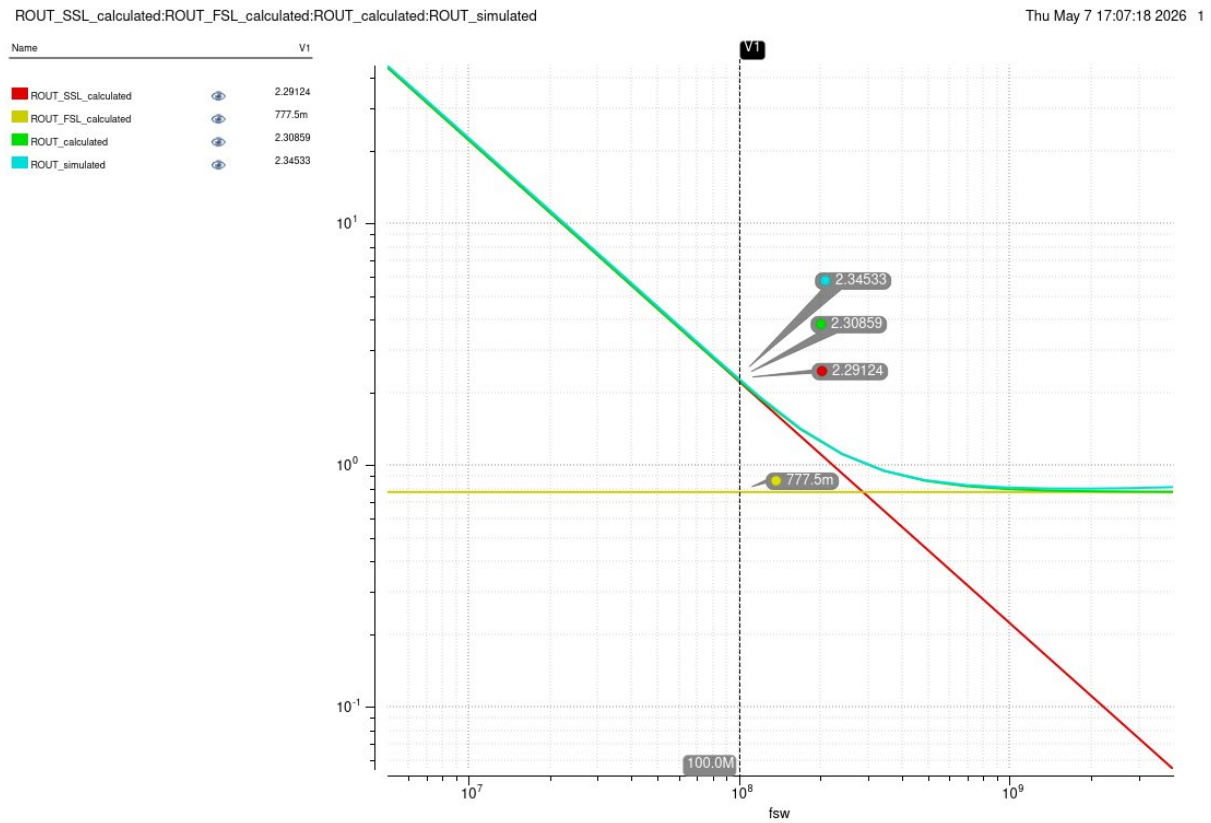
Name	Type	Details	Plot	Save	Spec
ROUT_SSI_calculated	expr	(0.2222 / VAR("R_f") / VAR("tau"))			
ROUT_FSL_calculated	expr	(1.555 * VAR("R_DSON"))			
ROUT_calculated	expr	(ROUT_SSI_calculated / tanh(ROUT_SSI_calculated / ROUT_FSL_calculated))			
ROUT_simulated	signal	/VOUT			
VIN	signal	/VIN			
VC1	signal	/VC1			
VC2	signal	/VC2			
VC2_bot	signal	/VC2_bot			
VOUT_average	expr	average(VT("VOUT"))			
VOUT_average_calculated_100MHz	expr	((VAR("VIN") / 3) + VAR("R_load") / (VAR("R_load") + 2.35))			
P_out	expr	(average(VT("VOUT"))**2) / VAR("R_load")			
P_cond	expr	(ROUT_calculated * (average(VT("VOUT")) / VAR("R_load"))**2)			
P_sw_bot	expr	(VAR("tau") * VAR("C_bot") + VAR("C_bot") / 12 * (average(VT("VOUT"))**2))			
P_sw_drain	expr	(VAR("tau") * VAR("C_drain") + 12 * (1**2) + 12 * (average(VT("VOUT"))**2))			
EFF	expr	(average(VT("VOUT"))**2) / VAR("R_load") / (average(VT("VOUT"))**2) / VAR("VIN") * 100)			
EFF_calculated	expr	(P_out / (P_out + P_cond + P_sw_bot + P_sw_drain)) * 100			
P_sw1	expr	(average(VT("VSW1_ABNUS")) / 2) * 1.8			
P_sw2	expr	(average(VT("VSW1_ABNUS")) / 2) * 2.5			
P_sw3	expr	(average(VT("VSW1_ABNUS")) / 2) * 2.5			
P_sw4	expr	(average(VT("VSW1_ABNUS")) / 2) * 1.8			
P_sw5	expr	(average(VT("VSW1_ABNUS")) / 2) * 1.8			
P_sw6	expr	(average(VT("VSW1_ABNUS")) / 2) * 3.1			
P_sw7	expr	(average(VT("VSW1_ABNUS")) / 2) * 3			
P_sw_gate	expr	(P_sw1 + P_sw2 + P_sw3 + P_sw4 + P_sw5 + P_sw6)			
EFF_Psw_gate	expr	(average(VT("VOUT"))**2) / VAR("R_load") / (average(VT("VOUT"))**2) + VAR("VIN") * 100)			
P_sw_gate_calculated	expr	(VAR("tau") * VAR("C_gate") * (1.8**2) + (2.5**2) + (2.5**2) + (1.8**2) + (1.8**2) + (3.1**2) + (3**2))			
EFF_Psw_gate_calculated	expr	(P_out / (P_out + P_cond + P_sw_bot + P_sw_drain + P_sw_gate_calculated)) * 100			
P_cond_simulated_normalized	expr	((VAR("VIN") / 3) - VOUT_average) * I_load / P_out			
P_sw_gate_simulated_normalized	expr	(P_sw_gate / P_out)			
P_simulated_SUM	expr	(P_cond_simulated_normalized + P_sw_gate_simulated_normalized)			
I_load	expr	(VOUT_average / VAR("R_load"))			

2. Output Resistance

To maintain consistency and facilitate a direct comparison with the Ladder topology, a similar color scheme is adopted throughout the simulation plots.

The simulation results, as highlighted in *Figure 4*, demonstrate high correlation with theoretical calculations, accurately reflecting the asymptotic behavior characterized by the Slow Switching Limit (*SSL*) and the Fast Switching Limit (*FSL*).

Additionally, to support subsequent analysis, a vertical marker is placed at 100 MHz to extract the necessary parameter values.

Figure 4: Comparison $ROUT_{simulated}$ and $ROUT_{calculated}$

3. Converter Waveforms

Key nodes and signals waveforms are shown for steady-state operation:

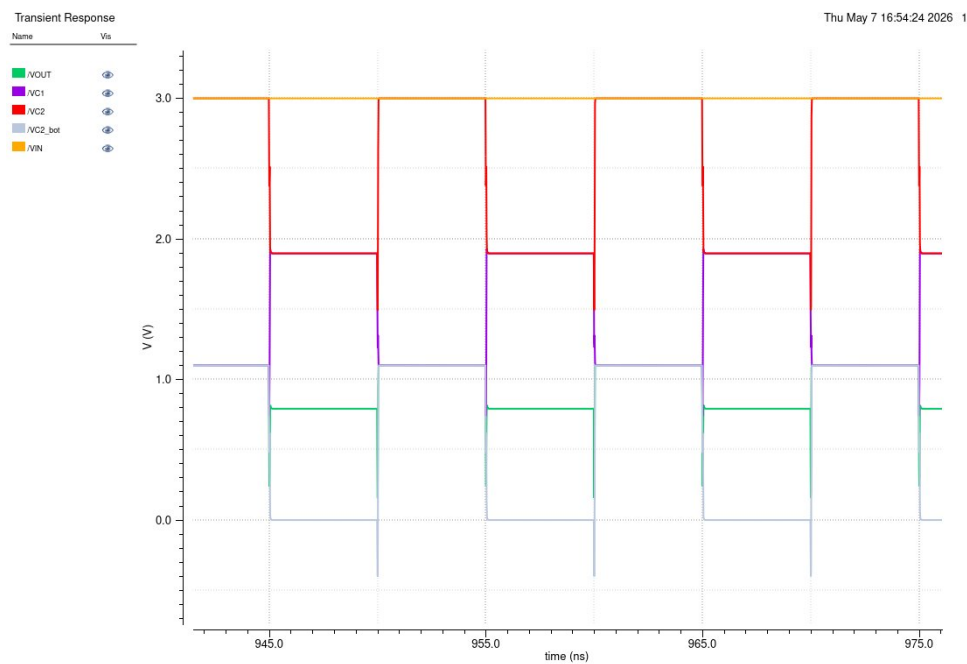


Figure 5: Waveforms of switching nodes and relevant signals

4. Converter Output Voltage

The discrepancies between the simulated and calculated output voltage waveforms across varying load resistances, observed in *Figure 6*, reveal the impact of physical parasitics on ideal converter models. At high R_{load} the simulated average V_{OUT} is lower than the analytically predicted value. This deviation is primarily attributed to parasitic capacitances at the switching nodes, specifically the bottom-plate capacitance k_{bott} and the intrinsic drain capacitances of the switches.

A comparison with the second simulation, clear in *Figure 7*, where k_{bott} is set to zero, shows a shift in the simulated curve toward the calculated expectation. However, a residual gap remains due to the parasitic drain capacitors of the switches.

This behavior confirms that the bottom-plate capacitance has a non-negligible contribution at high load resistances.

In the low R_{load} regime, there is a minor drift between simulated and calculated values. While the analytical model assumes a constant R_{out} characterized at a fixed $0.3V$ drop, the actual resistance varies slightly with the operating voltage drop ΔV . As load currents increase, this ΔV dependency introduces the observed discrepancy. Despite these minor deviations, the model remains accurate and fully capable of capturing the converter's overall performance.

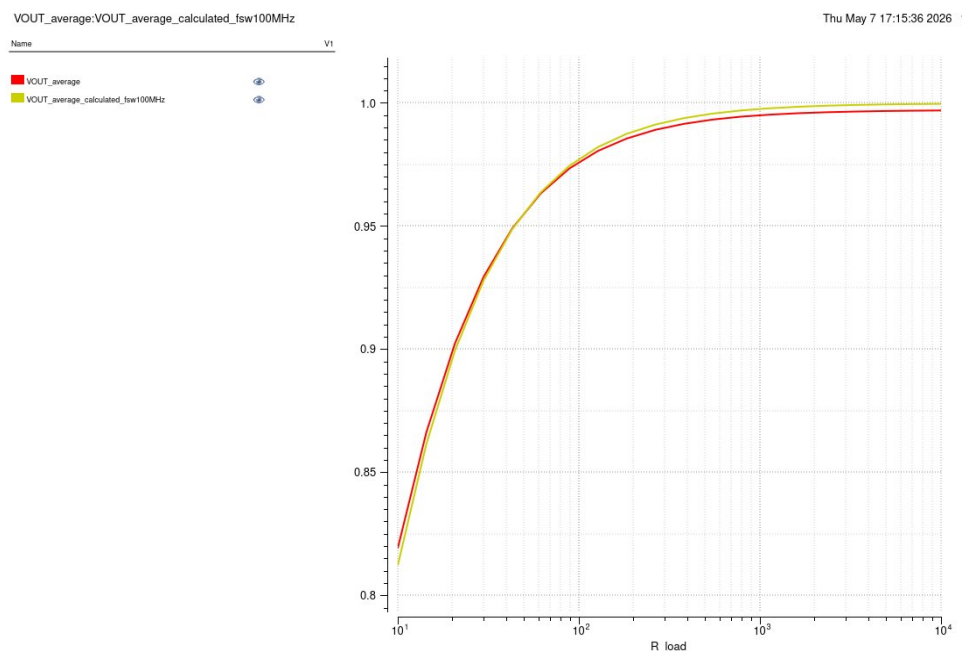


Figure 6: Output Voltage for $k_{bott} = 0.005$

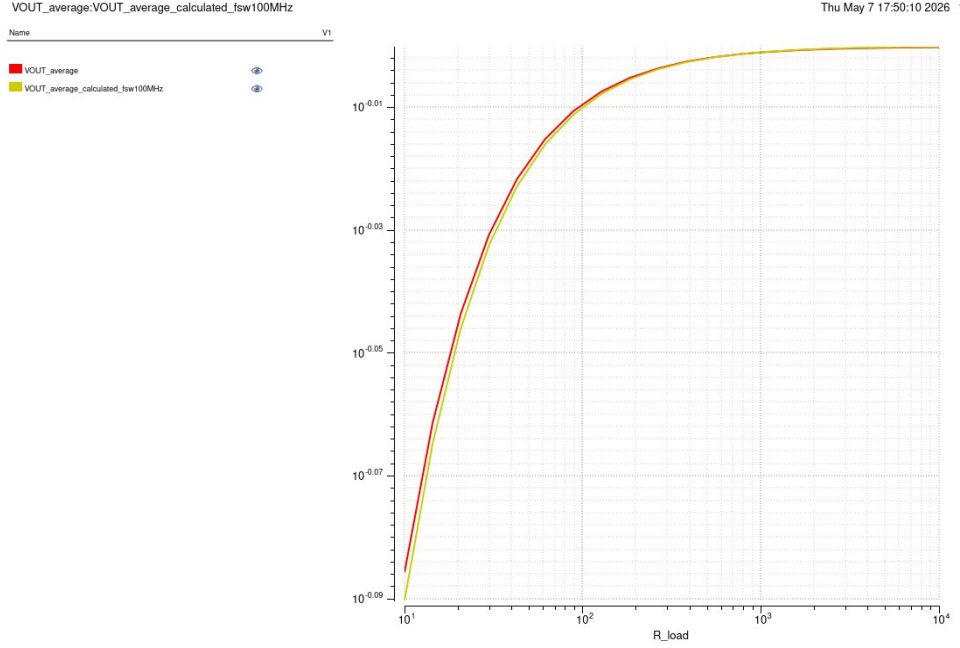


Figure 7: Output Voltage for ideal case ($k_{bott} = 0$)

5. Converter Efficiency neglecting the Gate Switching Losses

To calculate the converter efficiency highlighted in *Figure 8*, we need to properly model its losses.

Conduction Losses

The conduction losses are modeled through the R_{OUT} that we derived before, thus, they can be estimated as:

$$P_{COND} = R_{OUT} \cdot I_{OUT}^2 = R_{OUT} \cdot \left(\frac{V_{OUT}}{R_{load}} \right)^2$$

Bottom-Plate Switching Losses

The parasitic bottom-plate capacitors are switching at every cycle, thus, we can use the standard switching power loss equation:

$$P_{SW} = C_{bott} \cdot f_{sw} \cdot V_{swing}^2 = k_{bott} \cdot C_{fly} \cdot f_{sw} \cdot V_{swing}^2$$

However, to accurately estimate this contribution we need to know the voltage across the bottom-plate capacitors. From the schematic, we can see that the C_1 and C_2 bottom-plate switches between ground and V_{OUT} . Thus, we can estimate the switching losses as:

$$P_{SW,bott} = k_{bott} \cdot C_{fly} \cdot f_{sw} \cdot (2 * V_{OUT}^2)$$

Drain Switching losses

Similarly to the bottom-plate capacitors, the transistor drain parasitic capacitors are connected to the switching nodes, thus, their losses need to be estimated and added on top of the other contributions. The drain capacitances extracted from a separate testbench are $1.7pF$. Then, from the schematic it is possible to observe that SW_1 , as well as SW_4 , drain swing is equal to

V_{OUT} ; while SW_5 and SW_6 are nearly $1V$. The other drains are connected to nearly constant nodes and we can consider their losses to be negligible.

$$P_{SW,drain} = C_{drain} \cdot f_{sw} \cdot (2 * V_{OUT}^2 + 1^2 + 1^2)$$

Efficiency Calculation

Once the losses have been estimated, the efficiency can be calculated as follows:

$$\eta = \frac{P_{OUT}}{P_{IN}} = \frac{P_{OUT}}{P_{OUT} + P_{COND} + P_{SW,bott} + P_{SW,drain}} \cdot 100$$

Efficiency Simulation

As we are modeling a resistive R_{load} , the efficiency is calculated from the simulated results as follows:

$$\eta = \frac{P_{OUT}}{P_{IN}} = \frac{V_{OUT,average}^2 / R_{load}}{V_{IN} \cdot I_{IN,average}} \cdot 100$$

Neglecting switching losses, the maximum efficiency is achieved at a total output resistance of $R_{OUT} \approx 42 \Omega$ as well displayed in *Figure 8*.

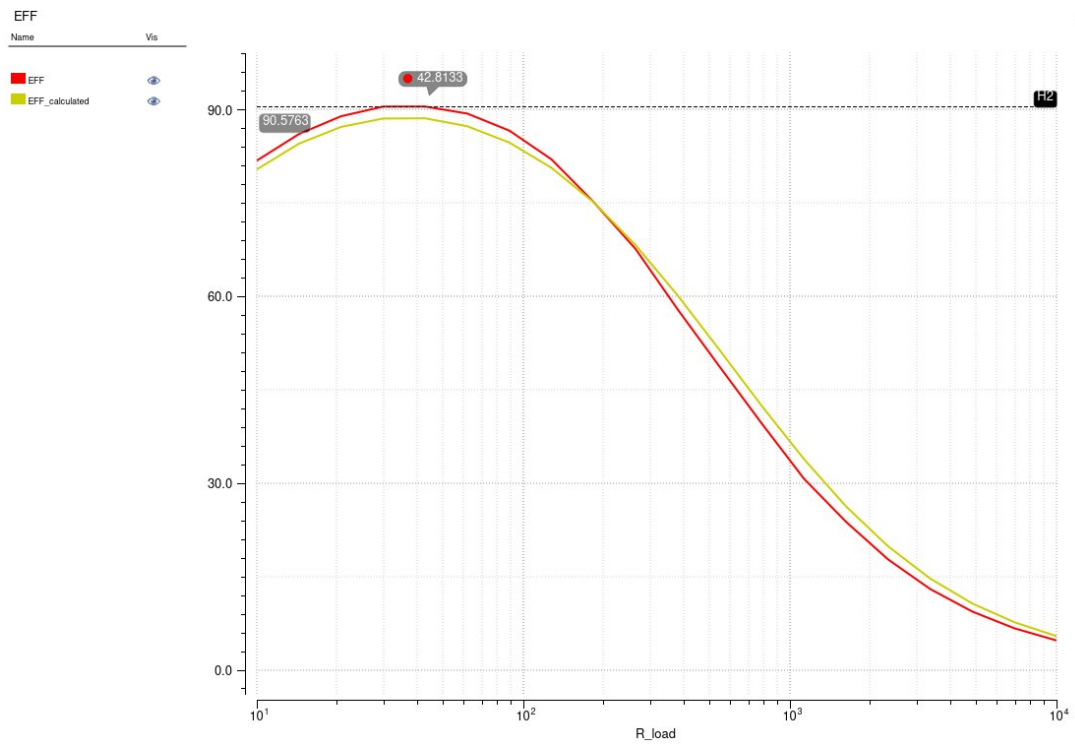


Figure 8: Converter Efficiency - no gate switching losses

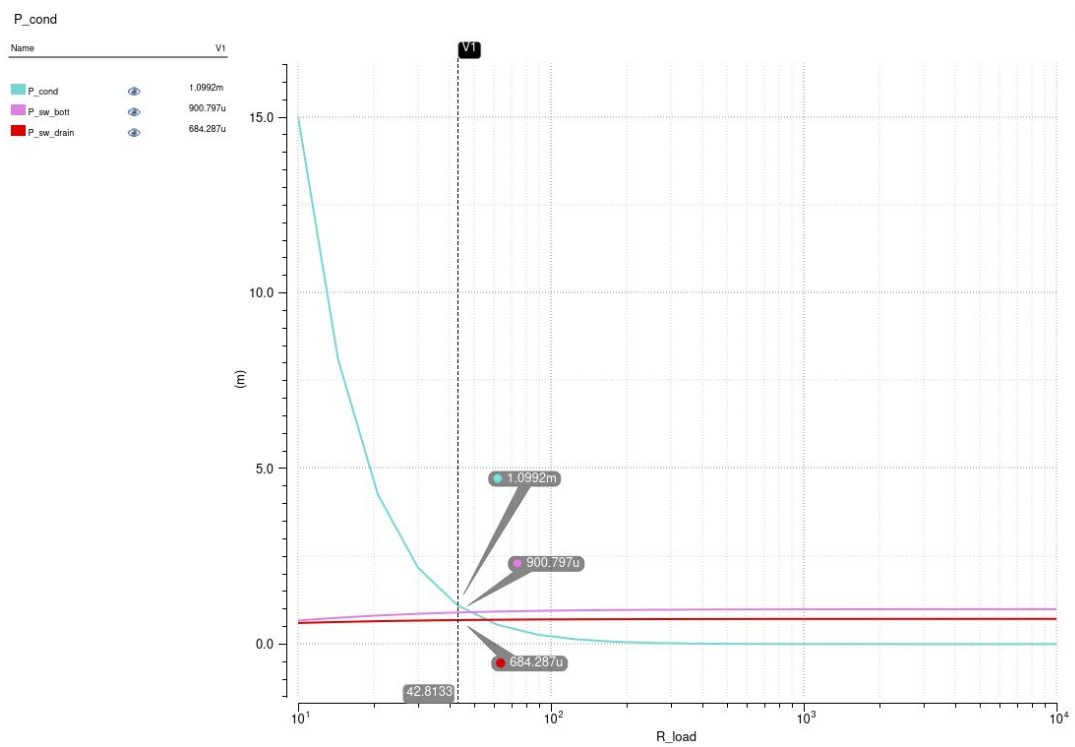


Figure 9: Comparison between Total Drain, Bottom-Plate and Conduction Losses

6. Converter Efficiency including the Gate Switching Losses

The gate switching losses can be calculated similarly to previous switching loss components by knowing the gate parasitic capacitance and the voltage swing applied to it.

The gate capacitance extraction leads to $C_{gate} = 3.1 \text{ pF}$.

$$P_{SW,gate} = C_{gate} \cdot f_{sw} \cdot V_{gate,swing}^2$$

The gate switching losses need to be added to the other loss contributions in the denominator of the efficiency equation.

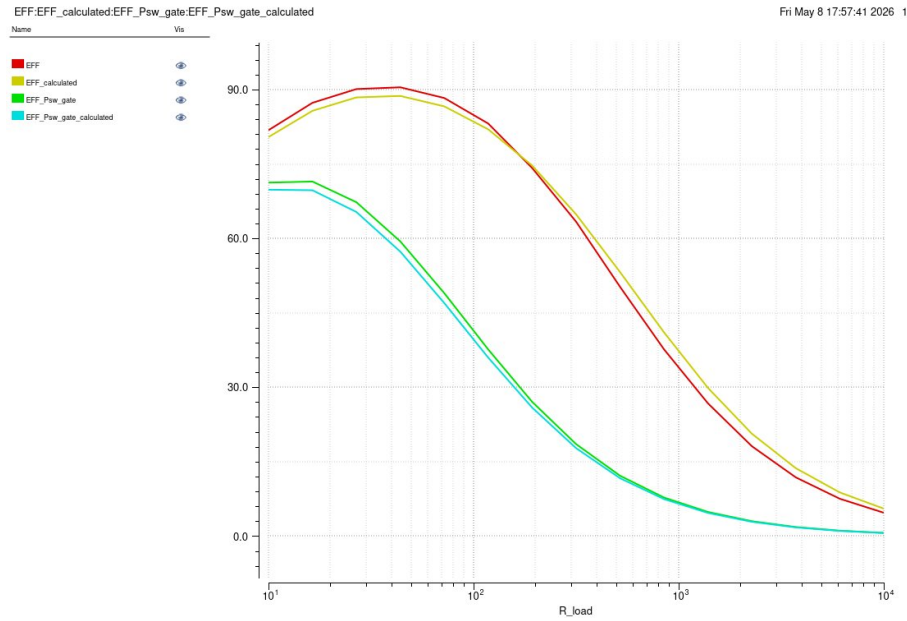


Figure 10: Converter Efficiency considering Gate Switching Losses

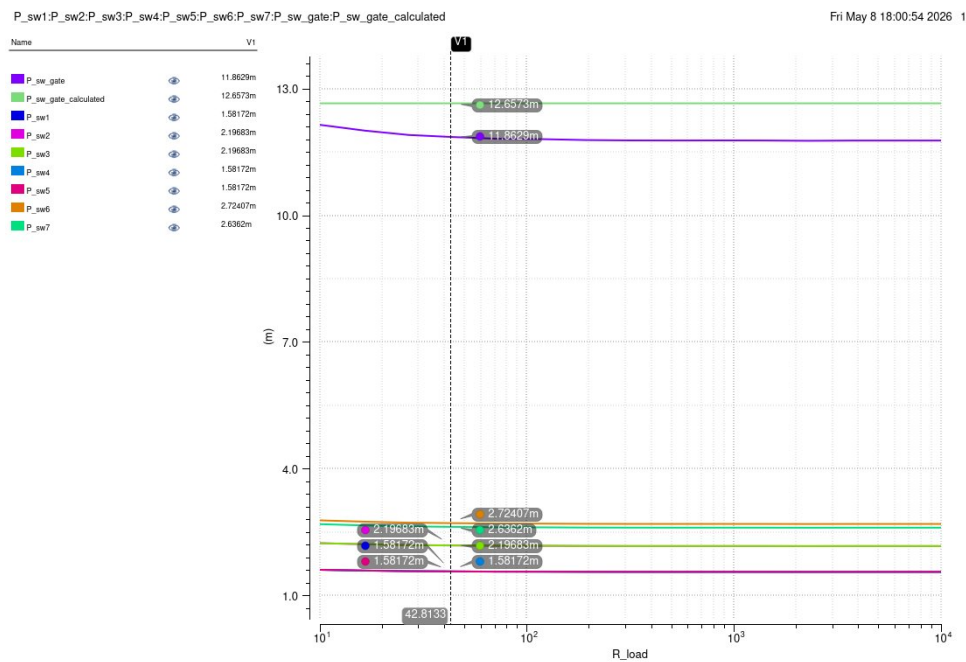


Figure 11: Gate Switching Losses

7. Losses normalized to Output Power

As illustrated in *Figure 12*, switching losses represent the dominant loss mechanism across the majority of the load range, significantly exceeding conduction losses in most operating conditions. The summation of these two loss components reveals a minimum point for a lower R_{load} compared to the one considering only the conduction related.

It is important to note that the analysis specific to this point neglects the contributions of bottom-plate and switch drain losses. Despite this, the results remain a robust representation of the converter's performance and effectively characterize the primary efficiency trends, since total C_{drain} and C_{bott} losses are one order of magnitude smaller than the one considered.

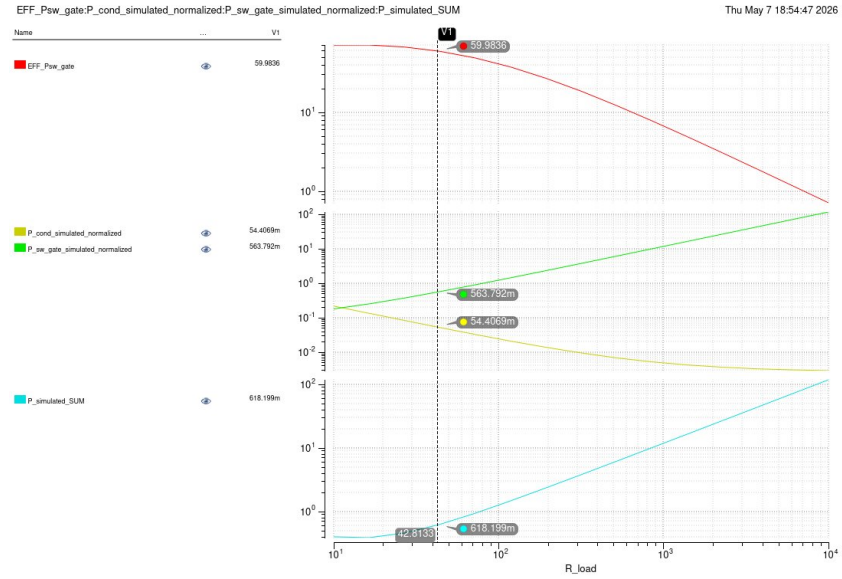


Figure 12: Normalized Output Power

8. Converter Efficiency Improvement

The fact that switching losses are significantly higher than conduction losses indicates a clear design opportunity to optimize the converter by reducing the switch size.

Reducing the switch width to $W' = W/2 = 47.5\mu\text{m}$ increases the $R_{DS,ON}$ from $500\text{m}\Omega$ to $900\text{m}\Omega$, which causes the conduction losses (P_{cond}) to increase proportionately. While this results in a higher conduction penalty, it serves as a strategic trade-off to address the dominant switching losses (P_{sw}). By halving the device width, the gate capacitance is also reduced by a factor of two, leading to a much smaller switching power requirement during each cycle.

Since the initial design was heavily limited by switching losses, the substantial reduction in P_{sw} far outweighs the rise in P_{cond} , as highlighted in *Figure 13*. Consequently, the sum of the normalized losses decreases across the load range, shifting the optimal operating point and resulting in a higher overall peak efficiency.

This outcome confirms that in regimes where switching losses are disproportionately high, increasing $R_{DS,ON}$ through smaller switch sizes is an effective strategy for maximizing total converter performance. The direct comparison between the previous case and the current one is clear in *Table 1*.

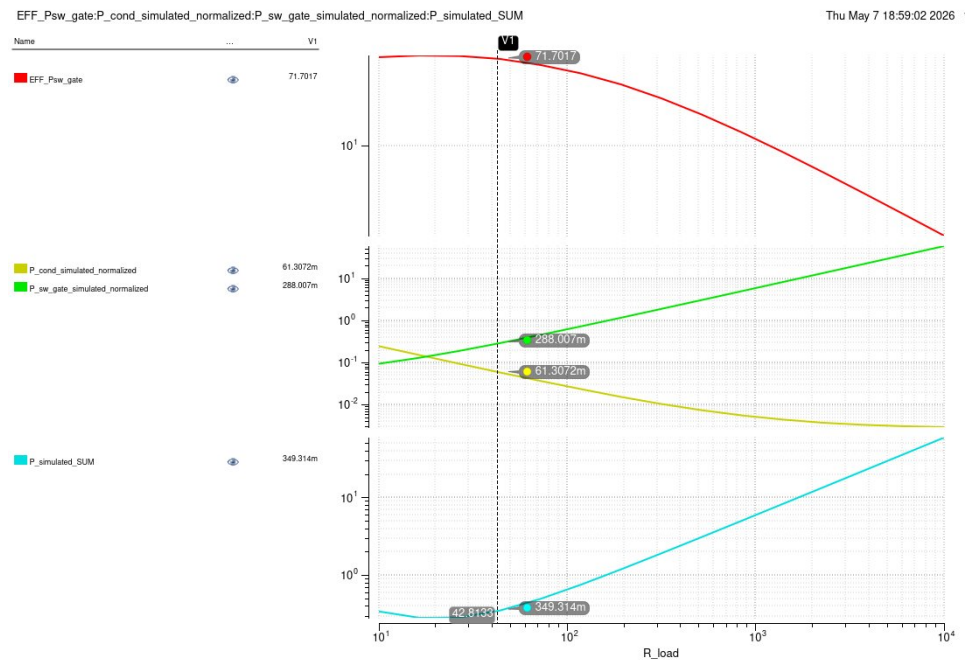


Figure 13: Normalized Output Power with $W/2$

Table 1: Comparison of Simulated Performance at $R_{load} \approx 42.8\Omega$

Parameter	Case 1 (W)	Case 2 (W/2)
Efficiency [%]	59.98	71.70
normalized P_{cond} [m]	54.41	61.31
normalized $P_{sw,gate}$ [m]	563.79	288.01
Total Loss (P_{sum}) [m]	618.20	349.31

9. Bonus Question

Designing all switches to be identical in a switched-capacitor converter typically leads to sub-optimal performance. Instead, the strategy proposed by the paper "*Analysis and Optimization of Switched Capacitor DC-DC Converters*" uses Lagrange multiplier optimization to size each switch individually based on its specific current and voltage requirements,

The Fast-Switching Limit (FSL) output impedance is mathematically defined as:

$$R_{FSL} = 2 \sum_i R_{DS_{ON},i} (a_{r,i})^2$$

To minimize conduction losses within a fixed silicon area constraint, the optimal conductance (G_i^*) for any given switch:

$$G_i^* = \frac{1}{R_i^*} \propto \frac{|a_{r,i}|}{v_{r,i}(\text{rated})}$$

Since a transistor's conductance scales with its physical width, this formula provides a clear sizing strategy. For a 3:1 Dickson converter, the process involves:

1. Calculating the charge multiplier ($a_{r,i}$) for each of the 7 switches.
2. Identifying the required blocking voltage ($v_{r,i}$) for each switch.
3. Scaling the area of each switch proportionally to the ratio $|a_{r,i}|/v_{r,i}$.

This approach ensures that switches handling the highest currents receive more die area to prevent bottlenecks, while those blocking higher voltages are sized to limit parasitic capacitive losses.

Implementation: Standard Switches ($S_1, S_2, S_3, S_4, S_5, S_7$): Since these all share the same blocking voltage (V_{OUT} or $1V$), they should all be sized with the same width, W_{base} .

The High-Stress Switch (S_6): Because this switch must block $2V$, it is effectively twice as "expensive" in terms of silicon area. To minimize the total output impedance R_{FSL} within your area budget, its width should be exactly half that of the others: $W_6 = W_{base}/2$.

Energy-efficient Analog Integrated Circuits

DCDC Converters – Assignment 2

Assignment

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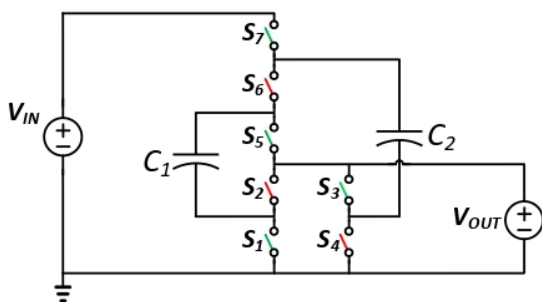
Introduction

In this assignment you will learn how to model and design two among the most popular Switched-Capacitor converter topologies and to understand what their benefits are for a certain set of design specifications. The first part of the Tutorial requires some hand calculation and a few plots of the derived equations, while the second part is done on cadence which will serve as a validation of the models as well as a design exercise.

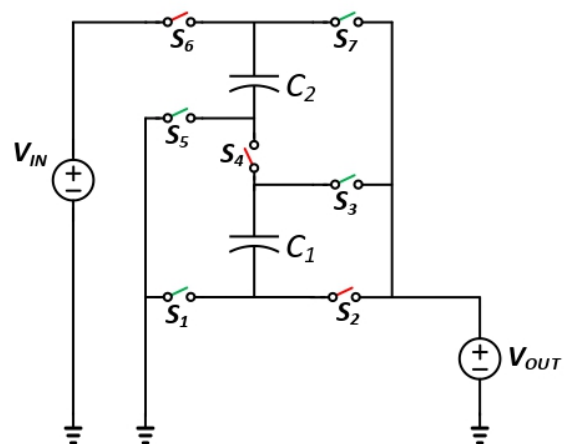
Part I – Modeling of the Dickson and Series-Parallel Switched-Capacitor Step-Down Converter

Given a Dickson and a Series-Parallel Switched-Capacitor topology with the same specifications as in the Tutorial:

Dickson



Series-Parallel



1. Draw circuit in phase 1 and phase 2
2. Calculate the voltage at which each flying capacitor will balance
3. Calculate the charge multiplier vectors for the slow switching limit R_{SSL}
4. Calculate the slow switching limit R_{SSL}
5. Calculate the charge multiplier vector for the fast switching limit R_{FSL}
6. Calculate the fast switching limit R_{FSL}
7. Calculate the converter Output Resistance
8. Write a comparison between the Ladder, Dickson and Series-parallel switched-capacitor converter topologies based on the following topics:
 - Conduction losses
 - Switching losses
 - Number of Flying Capacitors and Switches
 - Flying Capacitors and Switches utilization

Part II – Simulation of the Dickson Switched-Capacitor Converter on Cadence

1. Create the Dickson converter testbench taking inspiration from the ladder testbench used during the tutorial
 - a. Use ideal *vpulse* generators to control the switches and chose their swing to ensure a V_{gs} equal to 1.8V when the switch is closed and 0V when open, giving a 500mOhm $R_{DS(on)}$.
2. Run the Dickson converter testbench simulation and verify that the simulated output resistance matches the calculated one
3. Plot the converter time waveforms of the switching nodes and all the relevant signals
4. Simulate and calculate the converter output voltage
 - a. Choose an appropriate R_{load} range to observe a V_{OUT} drop down to 0.7V at the heaviest load
5. Calculate and Simulate the converter efficiency neglecting the gate switching losses
6. Calculate and Simulate the converter efficiency including the gate switching losses
7. Plot the simulated total conduction loss, the gate switching losses and their sum, all normalized to the output power
8. Can you suggest a design strategy to improve the converter efficiency?
9. Bonus Questions:
 - a. during all the previous analysis we have used all identical switches. However, as we are analysing step-down converters, when the voltage decreases the current increases. Therefore, we expect the current to be higher for the switches that are closer to the output. Could you devise a design strategy to optimize the converter efficiency sizing the switches accordingly to their current? Could you report the analysis and simulation results that motivate your strategy?